

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	XXXX
Project Title	HouseRent – House Rental Management System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **HouseRent – House Rental Management System** aims to simplify and modernize the property rental process through a secure web-based platform developed using the MERN stack. The system enables property owners to list and manage rental properties while allowing tenants to search, filter, and book houses online. It improves communication between tenants and property owners, reduces manual effort, and provides a secure and efficient rental management experience.

Project Overview	
Objective	To develop a secure, responsive, and user-friendly web application that simplifies property rental management by enabling tenants to search and book properties while allowing owners to manage listings efficiently.
Scope	The project covers user authentication, role-based access (Admin, Property Owner, Tenant), property listing and management, property search, booking management, and responsive dashboards using the MERN stack.
Problem Statement	
Description	Traditional property rental methods are time-consuming and inefficient. Tenants struggle to find verified rental properties, while property owners face challenges in managing listings and booking requests manually.
Impact	Implementing a centralized online rental management system will improve property discovery, simplify booking management, reduce manual work, and enhance the overall user experience for both tenants and property owners.
Proposed Solution	
Approach	Develop a full-stack MERN web application that provides secure authentication, property management, advanced search, and online booking through RESTful APIs and MongoDB.
Key Features	• Secure JWT-based User Authentication • Property Listing Management • Property Search & Advanced Filters • Online Property Booking System • Role-Based Access Control (Admin, Property Owner, Tenant)

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| | <ul style="list-style-type: none">• Responsive User Interface• Booking Dashboard for Owners and Tenants |
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Intel Core i3/i5 or above
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for application and database	Minimum 20 GB Free Storage (SSD Recommended)
Software		
Frontend Framework	User Interface Development	React.js, Vite, Bootstrap
Backend Framework	Server-Side Development	Node.js, Express.js
Database	Data Storage	MongoDB, Mongoose
Authentication	User Authentication & Security	JWT, bcrypt.js
Development Environment	IDE	Visual Studio Code
API Testing	REST API Testing	Postman
Version Control	Source Code Management	Git, GitHub
Data		
Data	Source, Type, Format	MongoDB Database, JSON/BSON, User, Property & Booking Data